

Stone Tao

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Education

PhD Student in Computer Science and Engineering

2023 –

University of California, San Diego. Advisor: Hao Su

Research Topics/Interests: Reinforcement Learning, Simulation, Embodied AI, Learning from Demonstrations

B.S. Double Major in Computer Science and Cognitive Science, Math Minor

2019 – 2023

University of California, San Diego.

Selected Awards

National Science Foundation Graduate Research Fellowship

2023-2028

UCSD CSE Award for Excellence in Leadership

2023

UCSD CSE Alumni Advisory Board Leadership Excellence Scholarship

2021-2022

Publications / Preprints

Staggered Environment Resets Improve Massively Parallel On-Policy Reinforcement Learning

Neural Information and Processing Systems (NeurIPS) 2025.

Sid Bharthulwar, **Stone Tao**, Hao Su.

[arXiv](#)

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

International Conference on Machine Learning (ICML) 2025.

Adrià López Escoriza, Nicklas Hansen, **Stone Tao**, Tongzhou Mu, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

Do Vision-Language Models Have Internal World Models? Towards an Atomic Evaluation

Association for Computational Linguistics (ACL) 2025.

Qiyue Gao*, Xinyu Pi*, Kevin Liu, Junrong Chen, Ruolan Yang, Xinqi Huang, Xinyu Fang, Lu Sun, Gautham Kishore, Bo Ai, **Stone Tao**, Mengyang Liu, Jiayi Yang, Chao-Jung Lai, Chuanyang Jin, Jiannan Xiang, Benhao Huang, Zeming Chen, David Danks, Hao Su, Tianmin Shu, Ziqiao Ma, Lianhui Qin, Zhiting Hu.

[arXiv](#) | [project page](#) | [code](#)

ManiSkill-HAB: A Benchmark for Low-Level Manipulation in Home Rearrangement Tasks

International Conference on Learning Representations (ICLR) 2025.

Arth Shukla, **Stone Tao**, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

Policy Decorator: Model-Agnostic Online Refinement for Large Policy Model

International Conference on Learning Representations (ICLR) 2025.

Xiu Yuan*, Tongzhou Mu*, **Stone Tao**, Yunhao Fang, Michael Zhang, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI

Robotics Science and Systems (RSS) 2025

Robot Learning Workshop at (ICLR) 2025, Oral Presentation

Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse-kai Chan, Yuan Gao, Xuanlin Li, Tongzhou Mu, Nan Xiao, Arnav Gurha, Viswesh Nagaswamy Rajesh, Yong Woo Choi, Yen-Ru Chen, Zhiao Huang, Roberto Calandra, Rui Chen, Shan Luo, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

Lux AI Season 3: Multi-Agent Meta Learning at Scale

Neural Information and Processing Systems (NeurIPS) 2024 Competitions Track.

Stone Tao, Akarsh Kumar, Bovard Doerschuk-Tiberi, Isabelle Pan, Addison Howard, Hao Su.

[competition page](#) | [code](#)

Reverse Forward Curriculum Learning for Extreme Sample and Demonstration Efficiency in RL

International Conference on Learning Representations (ICLR) 2024.

Stone Tao, Arth Shukla, Tse-kai Chan, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

MetaWriter: Exploring the Potential and Perils of AI Writing Support in Scientific Peer Review

26th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) 2024.

Lu Sun, Stone Tao, Junjie Hu, Steven Dow.

[ACM Digital Library Link](#)

Lux AI Season 2: NeurIPS Edition

Neural Information and Processing Systems (NeurIPS) 2023 Competitions Track.

Stone Tao, Qimai Li, Yuhao Jiang, Jiaxin Chen, Xiaolong Zhu, Bovard Doerschuk-Tiberi, Isabelle Pan, Addison Howard.

[competition page](#)

Abstract-to-Executable Trajectory Translation for One-Shot Task Generalization

International Conference on Machine Learning (ICML) 2023.

Stone Tao, Xiaochen Li, Tongzhou Mu, Zhiao Huang, Yuzhe Qin, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

ManiSkill2: a Unified Benchmark for Generalizable Manipulation Skills

International Conference on Learning Representations (ICLR) 2023

Jiayuan Gu[†], Fanbo Xiang[†], Xuanlin Li^{*}, Zhan Ling^{*}, Xiqiang Liu^{*}, Tongzhou Mu^{*}, Yihe Tang^{*}, Stone Tao^{*}, Xinyue Wei^{*}, Yunchao Yao^{*}, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

Maniskill: Generalizable manipulation skill benchmark with large-scale demonstrations

Neural Information and Processing Systems (NeurIPS) 2021 Datasets and Benchmarks Track

Tongzhou Mu^{*}, Zhan Ling^{*}, Fanbo Xiang^{*}, Derek Cathera Yang^{*}, Xuanlin Li^{*}, Stone Tao, Zhiao Huang, Zhiwei Jia, Hao Su.

[arXiv](#) | [project page](#) | [code](#)

Emergent Collective Intelligence From Massive-Agent Cooperation and Competition

Neural Information and Processing Systems (NeurIPS) 2022 Deep RL Workshop.

Hanmo Chen^{*}, Stone Tao^{*}, Jiaxin Chen, Weihang Shen, Xihui Li, Sikai Cheng, Xiaolong Zhu, Xiu Li.

[arXiv](#)

Experience

NVIDIA Research, Research Intern, Advisor: Iretiayo Akinola, Yijie Guo, Yashraj Narang 2025

- Research on sim2real, reinforcement learning, simulation, and manipulation.

Hillbot, Research Scientist Intern, Advisor: Hao Su 2024

- Research on the intersection of simulation, reinforcement learning, imitation learning, and embodied AI
- Business Strategy

UC San Diego: Hao Su Lab, Undergraduate Researcher, Advisor: Hao Su 2020 – 2023

- Research on robotics and embodied AI, particularly tackling long-horizon problems. Leveraging reinforcement learning, few-shot learning, imitation learning, and transformers.
- Research and develop large-scale robotic manipulation benchmarks (ManiSkill).

UC San Diego: ProtoLab, Undergraduate Researcher, Advisor: Steven Dow 2019 – 2023

- Research at the intersection of AI and HCI, particularly AI-assisted tools and using NLP in systems requiring in-depth synthesis of information such as peer-review systems.
- Developed the Design for San Diego competition's front and backend.

Lux AI Challenge, Co-Founder 2021 –

- Lead the research, development, and design of creative, accessible, and inclusive AI competitions.

- Past competitions have accumulated over 2000+ teams, 1,000+ github stars, 30,000+ submissions, 10,000,000+ matches run, and had one of the most diverse group of participants ever. Enabled competitors to research novel deep RL and IL methods to a unique large-scale multi-agent problem.

QuantCo, ML Engineer Intern, Advisor: T. Ben Thompson 2021

- Researched high-precision and accurate function approximation using neural nets and decision trees. Help automatically migrate slow, complex, hand-built computations in old systems to new systems.
- Developed OCR+NLP tools for analysis of insurance documents.

LaunchDarkly, Software Engineer Intern 2020

- Worked full stack on feature workflows, semantic patches, and conflict handling to enable state independent scheduling of feature flagging, a feature requested by LaunchDarkly's largest business customers.

Selected Open Source Projects

ManiSkill ★2000+ 2021 –

GPU parallelized robotics simulation framework with state-of-the-art simulation+rendering speeds and efficiency. Designed to be pythonic and abstract away most GPU memory management code to make building and training on robotics tasks a breeze.

[haosulab/ManiSkill](https://github.com/haosulab/ManiSkill)

Lux AI Challenge ★1600+ 2021 –

Code for the Lux AI Challenge, including the AI environments and visualizers.

[Lux-AI-Challenge](https://github.com/Lux-AI-Challenge)

Reinforcement Learning Gym and Library in Typescript ★20+ 2021

Implements a gym interface and algorithms like PPO and DQN in Typescript for reinforcement learning on browsers and Node.js.

[stonet2000/rl-ts](https://github.com/stonet2000/rl-ts)

Leadership

Co-Founder of the Lux AI Challenge 2021 –

Initiated the original project concept and invited others to co-found the non-profit organization and competition. Successfully some of the largest multi-agent AI competitions in collaboration with Kaggle. Raised over \$130,000 in funds for the competition.

Founding President of ACM AI at UC San Diego 2020 – 2022

Founded the largest organization on AI at UC San Diego, seeking to cultivate a community of AI enthusiasts as well as lower the barriers for learning AI. Grew the organization from zero to 1000+ members in 2 years, working with my board to develop social events, workshops, seminars, and competitions.

Presentations / Talks

Scaling Embodied AI via Simulation and Sim-integrated Robot Learning

Allen AI (AI2) 2025

Institute for AI Industry Research (AIR), Tsinghua University 2024

ManiSkill3: Scalable Simulation+Rendering for Generalizable Embodied AI

CVPR 2024 Embodied AI Workshop 2024

Echo AI Talk 2024

Lux AI Season 2 and Reinforcement Learning

Kaggle Podcast 2023

[Youtube recording](#)

AI Competition Design For Multi-Agent Environments / Games

Kaggle Podcast 2022

Learning in Foundation Environments (LIFE) 2022

[Slides](#)

Current/Past Student Mentees

Xiu Yuan (BS UC San Diego → PhD Washington University)	2024 – 2025
Xander Hinrichsen (BS UC San Diego → MS UC San Diego → NVIDIA)	2024 – 2025
Sid Bharthulwar (BS Harvard)	2024 – now
Rebecca Chen (BS UC San Diego)	2024 – now
Nan Xiao (MS UC San Diego → University of Tennessee, Knoxville)	2024 – now
Arnav Gurha (BS UC San Diego)	2024 – now
Arth Shukla (BS UC San Diego → PhD UC San Diego)	2023 – 2025
Kevin Chan (BS UC San Diego → MS Stanford)	2022 – 2024
Doris Zhang (BS UC San Diego)	2022 – 2024

Teaching

UC San Diego: CSE 276F Machine Learning for Robotics (Co-Instructor) 2024 Spring
Taught 4 lectures covering off policy RL, exploration in RL, learning from demonstrations, and model based RL. Wrote homework material covering robotics, robotics simulation, reinforcement learning, and robot learning. Held office hours.

UC San Diego: CSE 291 Machine Learning for Robotics 2023 Winter
Wrote various course materials and Google Colab notebooks covering robotics, robotics environments, reinforcement learning, and imitation learning.

UC San Diego: CSE 12 Basic Data Structures and Object-Oriented Design 2020 Spring
Held office hours and wrote unit tests for grading homework automatically in Java.

Service

Challenge Organization

Robotics: ManiSkill Challenge 2021 (ICLR 2022 Generalizable Policy Learning in the Physical World Workshop), ManiSkill2 Challenge (CVPR 2023 Embodied AI Workshop)
Multiagent Games/RL: Lux AI Season 1, Lux AI Season 2 (NeurIPS 2023 Competitions), Lux AI Season 3 (NeurIPS 2024 Competitions)

Peer Review

AI/ML: ICLR, ICML, NeurIPS (Top Reviewer), CoLLas, RLC
Robotics: IROS, ICRA
Computer Vision/Graphics: SIGGRAPH

Other Awards

UCSD Undergraduate Provost Honors List 2019-2023

MIT Battlecode (AI Competition) Finalist 2019-2021

Made finals 3 times in a row (2019-2021), placing 1st out of all soloists and 5th overall in 2020, competing against over 600 teams of high school to graduate students. Won the Five Rings adaptive strategy award for spearheading an influential strategy in 2021.

Skills

Programming Languages: Python, Typescript/Javascript, C/C++, Go, Java, SQL

AI: Reinforcement Learning, 2D/3D Computer Vision, Deep Learning, Tree Learners, Embodied AI

Frameworks: Jax, Pytorch, Tensorflow, Pandas, scikit-learn, Node.js, React, MongoDB, Express.js

Tools: Docker, Google Cloud, Jupyter Notebook/Lab, Git, Adobe Photoshop, Figma

Interests

Languages: English, Chinese

Sports: Fencing (Saber), Fencing Coaching

Music: Violin, Viola